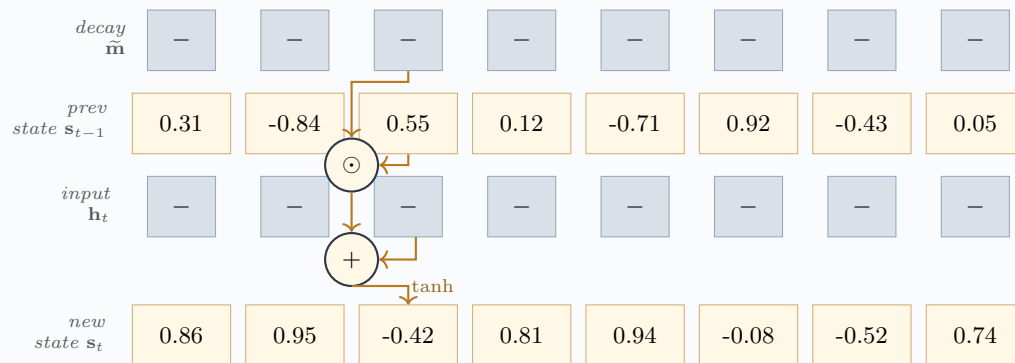


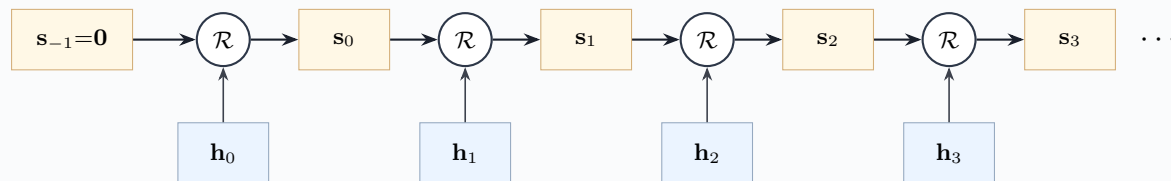
Block 3a. Recurrence — per-channel detail at one timestep t



channel $i=2$:
 $\tilde{m}_2 s_{t-1,2} + h_{t,2}$
 $= (+1)(0.55) + (-1)$
 $= -0.45$
 $\tanh(-0.45) \approx -0.42$

$$s_t = \tanh(\tilde{\mathbf{m}} \odot s_{t-1} + \mathbf{h}_t) \quad (\text{one channel at a time})$$

Block 3b. Recurrence — unrolled across time



$$\mathcal{R}(s_{t-1}, h_t) = \tanh(\tilde{\mathbf{m}} \odot s_{t-1} + \mathbf{h}_t) \quad O(d) \text{ per step. No attention. No key-value cache.}$$